

MACHINE LEARNING

HOMEWORK – 4

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Problem – 1

A.

We know,

k is a positive definite kernel function, such that:

$$k: R^d \times R^d \rightarrow R$$

By definition, a Positive Definite Kernel means that the Kernel Matrix (or Gram Matrix) is Positive Semi-definite.

Let the Kernel Matrix be denoted by K .

Thus, for a positive definite kernel,

$$x^T K x \geq 0 \tag{1}$$

$$\text{such that, } x = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ \vdots \\ c_n \end{pmatrix}, \text{ and } c_1, c_2, c_3, \dots, c_n \in R$$

Given,

$$k_a: R^d \times R^d \rightarrow R$$

$$k_a(x, z) = k(x, z) + 2$$

So,

$$k(z, x) = k(x, z)$$

$$k_a(z, x) = k(z, x) + 2$$

$$k_a(z, x) = k(x, z) + 2$$

$$\Rightarrow k_a(z, x) = k_a(x, z)$$

Thus, k_a is a symmetric kernel.

We know,

$$K_a = K + 2\mathbb{1}$$

Such that, $\mathbb{1}$ = all-ones matrix of dimensions $n \times n$

So,

$$\begin{aligned} x^T K_a x &= x^T (K + 2\mathbb{1}) x \\ &= x^T K x + x^T (2\mathbb{1}) x \\ &= x^T K x + 2(x^T \mathbb{1} x) \end{aligned} \tag{2}$$

We know,

$$\begin{aligned} x^T \mathbb{1} x &= \left(\sum_{i=1}^n c_i \right)^2 \\ \Rightarrow x^T \mathbb{1} x &\geq 0 \quad \forall x \in R^n \end{aligned} \tag{3}$$

Thus, from (1) and (3), we may conclude that,

$$\begin{aligned} x^T K x + 2(x^T \mathbb{1} x) &\geq 0 \quad \forall x \in R^n \\ \Rightarrow x^T K_a x &\geq 0 \quad \forall x \in R^n \end{aligned}$$

Thus,

K_a is positive semi-definite, and thus k_a is a positive definite kernel.

Therefore, **POSITIVE DEFINITE**

B.

We know,

k is a positive definite kernel function, such that:

$$k: R^d \times R^d \rightarrow R$$

By definition, a Positive Definite Kernel means that the Kernel Matrix (or Gram Matrix) is Positive Semi-definite.

Let the Kernel Matrix be denoted by K .

Thus, for a positive definite kernel,

$$x^T K x \geq 0 \quad (1)$$

$$\text{such that, } x = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ \vdots \\ c_n \end{pmatrix}, \text{ and } c_1, c_2, c_3, \dots, c_n \in \mathbb{R}$$

Given,

$$k_b: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$$

$$k_b(x, z) = 2k(x, z)$$

So, we know,

$$k(z, x) = k(x, z)$$

$$k_b(z, x) = 2k(z, x)$$

$$k_b(z, x) = 2k(x, z)$$

$$\Rightarrow k_b(z, x) = k_b(x, z)$$

Thus, k_b is a symmetric kernel.

We know,

$$K_b = 2K$$

So,

$$x^T K_b x = x^T (2K) x$$

$$= 2(x^T K x)$$

Using (1),

$$x^T K x \geq 0$$

$$\Rightarrow 2(x^T K x) \geq 0$$

$$\Rightarrow x^T K_b x \geq 0 \quad \forall x \in \mathbb{R}^n$$

Thus,

K_b is positive semi-definite, and thus k_b is a positive definite kernel.

Therefore, **POSITIVE DEFINITE**

C.

We know,

k and $\tilde{k}$ are positive definite kernel functions, such that:

$$k: R^d \times R^d \rightarrow R$$

$$\tilde{k}: R^d \times R^d \rightarrow R$$

By definition, a Positive Definite Kernel means that the Kernel Matrix (or Gram Matrix) is Positive Semi-definite.

Let the Kernel Matrix for kernel function k be denoted by K .

Let the Kernel Matrix for kernel function $\tilde{k}$ be denoted by $\tilde{K}$

Since, k and $\tilde{k}$ are positive definite kernel functions,

$$x^T K x \geq 0 \tag{1}$$

$$x^T \tilde{K} x \geq 0 \tag{2}$$

$$\text{such that, } x = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ \vdots \\ c_n \end{pmatrix}, \text{ and } c_1, c_2, c_3, \dots, c_n \in R$$

Given,

$$k_c: R^d \times R^d \rightarrow R$$

$$k_c(x, z) = k(x, z) + \tilde{k}(x, z)$$

So, we know,

$$k(z, x) = k(x, z), \text{ and } \tilde{k}(z, x) = \tilde{k}(x, z)$$

$$k_c(z, x) = k(z, x) + \tilde{k}(z, x)$$

$$\Rightarrow k_c(z, x) = k(x, z) + \tilde{k}(x, z)$$

$$\Rightarrow k_c(z, x) = k_c(x, z)$$

Thus, k_c is a symmetric kernel.

We know,

$$K_c = K + \tilde{K}$$

So,

$$\begin{aligned} x^T K_c x &= x^T (K + \tilde{K}) x \\ &= x^T K x + x^T \tilde{K} x \end{aligned}$$

Using (1) and (2),

$$x^T K x \geq 0, \text{ and } x^T \tilde{K} x \geq 0 \quad \forall x \in R^n$$

Thus,

$$\begin{aligned} x^T K x + x^T \tilde{K} x &\geq 0 \quad \forall x \in R^n \\ \Rightarrow x^T K_c x &\geq 0 \quad \forall x \in R^n \end{aligned}$$

Therefore,

K_c is positive semi-definite, and thus k_c is a positive definite kernel.

Therefore, **POSITIVE DEFINITE**

D.

We know,

k and $\tilde{k}$ are positive definite kernel functions, such that:

$$k: R^d \times R^d \rightarrow R$$

$$\tilde{k}: R^d \times R^d \rightarrow R$$

By definition, a Positive Definite Kernel means that the Kernel Matrix (or Gram Matrix) is Positive Semi-definite.

Let the Kernel Matrix for kernel function k be denoted by K .

Let the Kernel Matrix for kernel function $\tilde{k}$ be denoted by $\tilde{K}$

Since, k and $\tilde{k}$ are positive definite kernel functions,

$$x^T K x \geq 0 \tag{1}$$

$$x^T \tilde{K} x \geq 0 \quad (2)$$

$$\text{such that, } x = \begin{pmatrix} c_1 \\ c_2 \\ \dots \\ c_n \end{pmatrix}, \text{ and } c_1, c_2, c_3, \dots, c_n \in R$$

Given,

$$k_d: R^d \times R^d \rightarrow R$$

$$k_d(x, z) = k(x, z) - \tilde{k}(x, z)$$

So, we know,

$$k(z, x) = k(x, z), \text{ and } \tilde{k}(z, x) = \tilde{k}(x, z)$$

$$k_d(z, x) = k(z, x) - \tilde{k}(z, x)$$

$$\Rightarrow k_d(z, x) = k(x, z) - \tilde{k}(x, z)$$

$$\Rightarrow k_d(z, x) = k_d(x, z)$$

Thus, k_d is a symmetric kernel.

We know,

$$K_d = K - \tilde{K}$$

So,

$$x^T K_d x = x^T (K - \tilde{K}) x$$

$$= x^T K x - x^T \tilde{K} x$$

Using (1) and (2),

$$x^T K x \geq 0, \text{ and } x^T \tilde{K} x \geq 0 \quad \forall x \in R^n$$

However,

$$\text{If } x^T \tilde{K} x > x^T K x \quad \forall x \in R^n, \text{ then}$$

$$x^T K x - x^T \tilde{K} x < 0 \quad \forall x \in R^n$$

$$\Rightarrow x^T K_d x < 0 \quad \forall x \in R^n$$

Thus,

K_d is not necessarily positive semi-definite, and therefore k_b is not necessarily a positive definite kernel.

Therefore, **NOT NECESSARILY POSITIVE DEFINITE**

Problem – 2

We know,

A is some $n \times p$ matrix, and $\lambda > 0$

To prove:

$$(A^T A + \lambda I)^{-1} A^T = A^T (A A^T + \lambda I)^{-1}$$

Proving the above identity is essentially the same as proving the following:

$$\Rightarrow A^T (A A^T + \lambda I) = (A^T A + \lambda I) A^T \quad (1)$$

Solving the Left-Hand Side of (1),

$$= A^T (A A^T + \lambda I)$$

$$= A^T (A A^T) + A^T \lambda I$$

Since, λ is just a scalar quantity,

$$= A^T (A A^T) + \lambda (A^T I) \quad (2)$$

Using associative property of multiplication of matrices,

$$A^T (A A^T) = (A^T A) A^T$$

Substituting the above relation in (2),

$$= (A^T A) A^T + \lambda (A^T I) \quad (3)$$

Here, the identity matrix I is of dimensions $n \times n$

$$= (A^T A) A^T + \lambda (A^T I_n) \quad (4)$$

We know that,

$$A^T I_n = I_p A^T = A^T$$

Therefore, (4) is the same as:

$$= (A^T A) A^T + \lambda (I_p A^T) \quad (5)$$

Solving the Right-Hand Side of (1),

$$= (A^T A + \lambda I) A^T$$

$$= (A^T A) A^T + \lambda I A^T$$

Since, λ is just a scalar quantity,

$$= (A^T A) A^T + \lambda (I A^T)$$

$$= (A^T A) A^T + \lambda (I_p A^T) \quad (6)$$

Using (5) and (6), we see that the Left-Hand Side of (1) = Right-Hand Side of (1)

$$\Rightarrow A^T (A A^T + \lambda I) = (A^T A + \lambda I) A^T$$

$$\Rightarrow (A^T A + \lambda I)^{-1} A^T = A^T (A A^T + \lambda I)^{-1}$$

Hence, proved.

Problem – 3

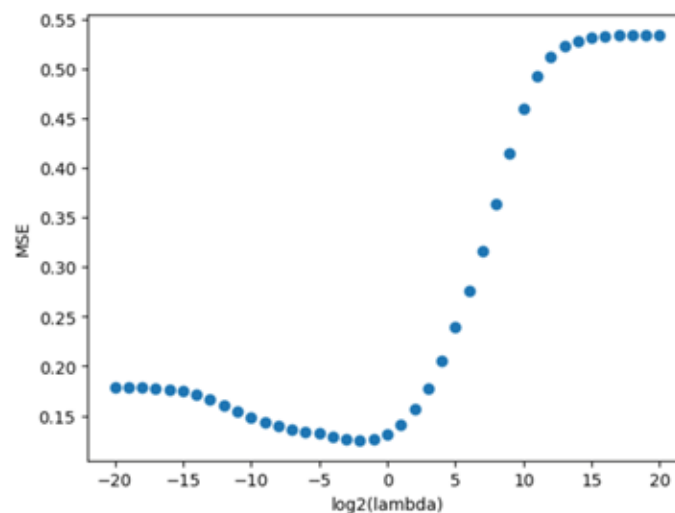
A.

Since the integrand in the hint given by the kernel is always positive then this kernel function is always positive, and therefore the kernel matrix will always be semi-positive definite which makes this a positive definite kernel!

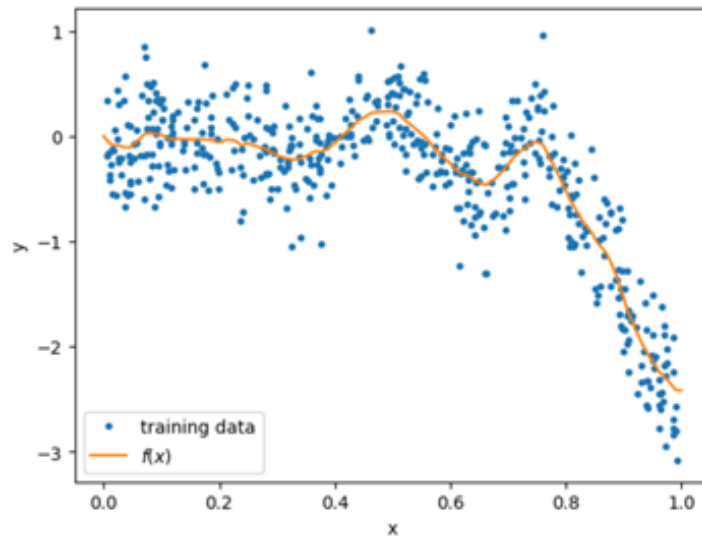
B.

The lambda of smallest mean squared error on the validation set is:

$$\lambda = 2^{-2} \text{ with } MSE = 0.12519$$



The final MSE on the test set was 0.109495, and the plot of the regression fit over the training data using the lambda found by the validations is:



Problem – 4

A.

TRUE

q_1 and q_2 are orthonormal vectors in R^d , such that the subspace spanned by $\{Aq_1, Aq_2\}$ is the same as the subspace spanned by $\{u_1, u_2\}$.

This implies that the columns of Aq_1 and Aq_2 span the same subspace as the columns of u_1 and u_2 . Thus, the column space of Aq_1 and Aq_2 is the same as the column space of u_1 and u_2 .

The singular value decomposition of A is given by:

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T$$

In matrix form, we know that,

$$A = U \Sigma V^T$$

Where,

columns of U – left singular vectors of A

columns of V – right singular vectors of A

Σ – diagonal matrix containing $\sigma_1, \sigma_2, \dots, \sigma_r$ (in decreasing order)

The subspace spanned by $\{u_1, u_2\}$ correspond to the column space of U.

$$\Rightarrow \text{span}\{u_1, u_2\} = CS(U)$$

The subspace spanned by $\{Aq_1, Aq_2\}$ correspond to the column space of AV.

$$\Rightarrow \text{span}\{Aq_1, Aq_2\} = CS(V)$$

We know that,

If the subspace spanned by $\{Aq_1, Aq_2\}$ is the same as the subspace spanned by $\{u_1, u_2\}$.

This implies that the columns of U corresponding to u_1 , and u_2 are the same as the columns of V corresponding to Aq_1 , and Aq_2 .

Thus,

$$u_1 = V(Aq_1)$$

$$u_2 = V(Aq_2)$$

Thus, calculating the norms of u_1 , and u_2 , we obtain the following:

$$\|u_1\|^2 = \|V(Aq_1)\|^2$$

$$\|u_2\|^2 = \|V(Aq_2)\|^2$$

Since V is an orthogonal matrix, the norms of it's columns are preserved.

Therefore,

$$\|V(Aq_1)\|^2 = \|Aq_1\|^2$$

$$\|V(Aq_2)\|^2 = \|Aq_2\|^2$$

Thus,

$$\|u_1\|^2 = \|Aq_1\|^2$$

$$\|u_2\|^2 = \|Aq_2\|^2$$

Since the subspace spanned by $\{Aq_1, Aq_2\}$ is the same as the subspace spanned by $\{u_1, u_2\}$.

$$\Rightarrow \|u_1\|^2 + \|u_2\|^2 = \|Aq_1\|^2 + \|Aq_2\|^2$$

$$\Rightarrow \|Aq_1\|^2 + \|Aq_2\|^2 = \|u_1\|^2 + \|u_2\|^2$$

B.

FALSE

As given,

$$\|Aq_1\|^2 + \|Aq_2\|^2 = \sigma_1^2 + \sigma_2^2$$

The above relation suggests that the sum of the squared norms of Aq_1 and Aq_2 is equal to the sum of the squared singular values (σ_1^2 and σ_2^2).

However, it does not necessarily imply that the subspace spanned by $\{Aq_1, Aq_2\}$ is the same as the subspace spanned by $\{u_1, u_2\}$.

The singular value decomposition of A is given by:

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T$$

In matrix form, we know that,

$$A = U \Sigma V^T$$

Where,

columns of U – left singular vectors of A

columns of V – right singular vectors of A

Σ – diagonal matrix containing $\sigma_1, \sigma_2, \dots, \sigma_r$ (in decreasing order)

The subspace spanned by $\{u_1, u_2\}$ correspond to the column space of U .

$$\Rightarrow \text{span}\{u_1, u_2\} = CS(U)$$

The subspace spanned by $\{Aq_1, Aq_2\}$ correspond to the column space of AV .

$$\Rightarrow \text{span}\{Aq_1, Aq_2\} = CS(V)$$

The equality provided in the problem only equates the norms of the transformed vectors Aq_1 , and Aq_2 to the singular values σ_1 , and σ_2 .

It does not guarantee that the subspaces spanned by $\{Aq_1, Aq_2\}$, and $\{u_1, u_2\}$ are the same, unless additional information or constraints are specified.

Problem – 5

In this chart the row and columns are labeled by the indices of the words in the 'words_to_compare'

The mapping goes like:

excellent = 0
amazing = 1
delicious = 2
fantastic = 3
gem = 4
perfectly = 5
incredible = 6
worst = 7
mediocre = 8
bland = 9
meh = 10
awful = 11
horrible = 12
terrible = 13

In this chart the row and columns are labeled by the indices of the words in the 'words_to_compare'

The mapping goes like:
 excellent = 0
 amazing = 1
 delicious = 2
 fantastic = 3
 gem = 4
 perfectly = 5
 incredible = 6
 worst = 7
 mediocre = 8
 bland = 9
 meh = 10
 awful = 11
 horrible = 12
 terrible = 13

k = 2

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	1.00	0.85	0.85	0.96	0.97	0.99	0.98	0.40	0.59	0.79	0.13	0.33	0.24	0.55
1	0.85	1.00	1.00	0.97	0.96	0.90	0.93	0.82	0.92	0.99	0.63	0.77	0.71	0.91
2	0.85	1.00	1.00	0.97	0.96	0.90	0.93	0.82	0.93	1.00	0.63	0.78	0.72	0.91
3	0.96	0.97	0.97	1.00	1.00	0.98	0.99	0.64	0.80	0.94	0.41	0.59	0.51	0.77
4	0.97	0.96	0.96	1.00	1.00	0.99	1.00	0.62	0.78	0.92	0.38	0.56	0.48	0.75
5	0.99	0.90	0.90	0.98	0.99	1.00	1.00	0.49	0.67	0.85	0.23	0.42	0.34	0.63
6	0.98	0.93	0.93	0.99	1.00	1.00	1.00	0.56	0.73	0.89	0.31	0.50	0.42	0.70
7	0.40	0.82	0.82	0.64	0.62	0.49	0.56	1.00	0.98	0.87	0.96	1.00	0.99	0.98
8	0.59	0.92	0.93	0.80	0.78	0.67	0.73	0.98	1.00	0.96	0.88	0.96	0.93	1.00
9	0.79	0.99	1.00	0.94	0.92	0.85	0.89	0.87	0.96	1.00	0.70	0.83	0.78	0.94
10	0.13	0.63	0.63	0.41	0.38	0.23	0.31	0.96	0.88	0.70	1.00	0.98	0.99	0.90
11	0.33	0.77	0.78	0.59	0.56	0.42	0.50	1.00	0.96	0.83	0.98	1.00	1.00	0.97
12	0.24	0.71	0.72	0.51	0.48	0.34	0.42	0.99	0.93	0.78	0.99	1.00	1.00	0.94
13	0.55	0.91	0.91	0.77	0.75	0.63	0.70	0.98	1.00	0.94	0.90	0.97	0.94	1.00

k = 4

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	1.00	0.84	0.83	0.88	0.37	0.90	0.91	0.24	0.40	0.34	0.04	0.22	0.22	0.39
1	0.84	1.00	0.93	0.92	0.50	0.75	0.92	0.34	0.55	0.33	0.35	0.30	0.23	0.40
2	0.83	0.93	1.00	0.89	0.38	0.74	0.79	0.42	0.52	0.31	0.26	0.44	0.53	0.66
3	0.88	0.92	0.89	1.00	0.72	0.64	0.95	0.02	0.21	0.01	-0.05	0.01	0.11	0.26
4	0.37	0.50	0.38	0.72	1.00	-0.03	0.65	-0.62	-0.43	-0.62	-0.47	-0.63	-0.43	-0.37
5	0.90	0.75	0.74	0.64	-0.03	1.00	0.73	0.59	0.71	0.71	0.39	0.54	0.38	0.55
6	0.91	0.92	0.79	0.95	0.65	0.73	1.00	0.04	0.28	0.15	0.04	-0.02	-0.06	0.11
7	0.24	0.34	0.42	0.02	-0.62	0.59	0.04	1.00	0.95	0.92	0.87	0.98	0.75	0.80
8	0.40	0.55	0.52	0.21	-0.43	0.71	0.28	0.95	1.00	0.94	0.92	0.89	0.57	0.68
9	0.34	0.33	0.31	0.01	-0.62	0.71	0.15	0.92	0.94	1.00	0.83	0.85	0.48	0.58
10	0.04	0.35	0.26	-0.05	-0.47	0.39	0.04	0.87	0.92	0.83	1.00	0.79	0.41	0.48
11	0.22	0.30	0.44	0.01	-0.63	0.54	-0.02	0.98	0.89	0.85	0.79	1.00	0.86	0.89
12	0.22	0.23	0.53	0.11	-0.43	0.38	-0.06	0.75	0.57	0.48	0.41	0.86	1.00	0.98
13	0.39	0.40	0.66	0.26	-0.37	0.55	0.11	0.80	0.68	0.58	0.48	0.89	0.98	1.00

k = 8

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	1.00	0.76	0.91	0.95	0.64	0.73	0.89	-0.45	-0.25	0.20	-0.06	-0.41	-0.29	-0.32
1	0.76	1.00	0.89	0.87	0.46	0.31	0.80	-0.10	0.01	0.29	0.12	-0.01	0.15	0.06
2	0.91	0.89	1.00	0.96	0.60	0.65	0.87	-0.44	-0.17	0.31	0.10	-0.27	-0.18	-0.25
3	0.95	0.87	0.96	1.00	0.74	0.58	0.88	-0.46	-0.24	0.16	0.01	-0.40	-0.24	-0.28
4	0.64	0.46	0.60	0.74	1.00	0.37	0.69	-0.62	-0.54	-0.32	-0.27	-0.70	-0.51	-0.52
5	0.73	0.31	0.65	0.58	0.37	1.00	0.69	-0.65	-0.24	0.42	0.07	-0.39	-0.59	-0.60
6	0.89	0.80	0.87	0.88	0.69	0.69	1.00	-0.40	-0.28	0.17	-0.13	-0.32	-0.25	-0.33
7	-0.45	-0.10	-0.44	-0.46	-0.62	-0.65	-0.40	1.00	0.64	0.01	0.08	0.80	0.90	0.92
8	-0.25	0.01	-0.17	-0.24	-0.54	-0.24	-0.28	0.64	1.00	0.68	0.79	0.86	0.60	0.72
9	0.20	0.29	0.31	0.16	-0.32	0.42	0.17	0.01	0.68	1.00	0.85	0.50	0.08	0.14
10	-0.06	0.12	0.10	0.01	-0.27	0.07	-0.13	0.08	0.79	0.85	1.00	0.50	0.12	0.24
11	-0.41	-0.01	-0.27	-0.40	-0.70	-0.39	-0.32	0.80	0.86	0.50	0.50	1.00	0.83	0.86
12	-0.29	0.15	-0.18	-0.24	-0.51	-0.59	-0.25	0.90	0.60	0.08	0.12	0.83	1.00	0.98
13	-0.32	0.06	-0.25	-0.28	-0.52	-0.60	-0.33	0.92	0.72	0.14	0.24	0.86	0.98	1.00

Problem – 6

The mapping goes like:

excellent = 0
amazing = 1
delicious = 2
fantastic = 3
gem = 4
perfectly = 5
incredible = 6
worst = 7
mediocre = 8
bland = 9
meh = 10
awful = 11
horrible = 12
terrible = 13

After getting rid of the vector corresponding to the largest singular value, we noticed that the associations become much more intuitive and accurate. The positive terms are much more similar to each other and further away from the negative words and vice versa for the negative words as well. We would think that this is because the first singular vector brings everything closer together than what they intuitively should be.

In this chart the row and columns are labeled by the indices of the words in the 'words_to_compare'
The mapping goes like:

```

excellent = 0
amazing = 1
delicious = 2
fantastic = 3
gem = 4
perfectly = 5
incredible = 6
worst = 7
mediocre = 8
bland = 9
meh = 10
awful = 11
horrible = 12
terrible = 13

```

PROBLEM 6

the outputs after removing the first singular vector are:

k = 2	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	1.00	0.94	0.98	0.85	0.21	0.91	0.87	-0.04	0.14	0.30	-0.50	-0.06	-0.08	0.09
1	0.94	1.00	0.85	0.98	0.53	0.72	0.99	-0.38	-0.20	-0.05	-0.77	-0.40	-0.41	-0.26
2	0.98	0.85	1.00	0.73	0.00	0.98	0.75	0.17	0.34	0.49	-0.31	0.14	0.13	0.29
3	0.85	0.98	0.73	1.00	0.69	0.56	1.00	-0.55	-0.39	-0.24	-0.88	-0.57	-0.59	-0.44
4	0.21	0.53	0.00	0.69	1.00	-0.21	0.66	-0.98	-0.94	-0.87	-0.95	-0.99	-0.99	-0.96
5	0.91	0.72	0.98	0.56	-0.21	1.00	0.59	0.38	0.54	0.66	-0.10	0.35	0.34	0.49
6	0.87	0.99	0.75	1.00	0.66	0.59	1.00	-0.52	-0.36	-0.21	-0.86	-0.54	-0.56	-0.41
7	-0.04	-0.38	0.17	-0.55	-0.98	0.38	-0.52	1.00	0.98	0.94	0.88	1.00	1.00	0.99
8	0.14	-0.20	0.34	-0.39	-0.94	0.54	-0.36	0.98	1.00	0.99	0.79	0.98	0.98	1.00
9	0.30	-0.05	0.49	-0.24	-0.87	0.66	-0.21	0.94	0.99	1.00	0.68	0.93	0.93	0.98
10	-0.50	-0.77	-0.31	-0.88	-0.95	-0.10	-0.86	0.88	0.79	0.68	1.00	0.90	0.90	0.82
11	-0.06	-0.40	0.14	-0.57	-0.99	0.35	-0.54	1.00	0.98	0.93	0.90	1.00	1.00	0.99
12	-0.08	-0.41	0.13	-0.59	-0.99	0.34	-0.56	1.00	0.98	0.93	0.90	1.00	1.00	0.99
13	0.09	-0.26	0.29	-0.44	-0.96	0.49	-0.41	0.99	1.00	0.98	0.82	0.99	0.99	1.00

k = 4	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	1.00	0.95	0.93	0.95	0.60	0.96	0.96	-0.79	-0.62	0.40	-0.15	-0.70	-0.69	-0.69
1	0.95	1.00	0.98	0.99	0.69	0.94	0.95	-0.93	-0.77	0.34	-0.08	-0.84	-0.83	-0.85
2	0.93	0.98	1.00	0.96	0.62	0.93	0.88	-0.90	-0.75	0.36	-0.04	-0.77	-0.74	-0.77
3	0.95	0.99	0.96	1.00	0.78	0.89	0.96	-0.94	-0.83	0.19	-0.25	-0.86	-0.81	-0.82
4	0.60	0.69	0.62	0.78	1.00	0.43	0.77	-0.84	-0.94	-0.43	-0.65	-0.92	-0.75	-0.75
5	0.96	0.94	0.93	0.89	0.43	1.00	0.88	-0.76	-0.52	0.61	0.13	-0.63	-0.70	-0.71
6	0.96	0.95	0.88	0.96	0.77	0.88	1.00	-0.87	-0.75	0.22	-0.27	-0.85	-0.82	-0.81
7	-0.79	-0.93	-0.90	-0.94	-0.84	-0.76	-0.87	1.00	0.93	-0.05	0.18	0.97	0.92	0.94
8	-0.62	-0.77	-0.75	-0.83	-0.94	-0.52	-0.75	0.93	1.00	0.30	0.46	0.94	0.79	0.82
9	0.40	0.34	0.36	0.19	-0.43	0.61	0.22	-0.05	0.30	1.00	0.78	0.11	-0.14	-0.14
10	-0.15	-0.08	-0.04	-0.25	-0.65	0.13	-0.27	0.18	0.46	0.78	1.00	0.30	0.02	-0.01
11	-0.70	-0.84	-0.77	-0.86	-0.92	-0.63	-0.85	0.97	0.94	0.11	0.30	1.00	0.94	0.95
12	-0.69	-0.83	-0.74	-0.81	-0.75	-0.70	-0.82	0.92	0.79	-0.14	0.02	0.94	1.00	0.99
13	-0.69	-0.85	-0.77	-0.82	-0.75	-0.71	-0.81	0.94	0.82	-0.14	-0.01	0.95	0.99	1.00

k = 8	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	1.00	0.83	0.80	0.96	0.76	0.43	0.83	-0.35	0.16	0.24	-0.03	-0.22	0.03	0.01
1	0.83	1.00	0.82	0.90	0.61	0.12	0.79	-0.09	0.27	0.30	0.10	0.07	0.31	0.24
2	0.80	0.82	1.00	0.88	0.57	0.56	0.86	-0.52	-0.15	0.28	0.04	-0.32	-0.14	-0.22
3	0.96	0.90	0.88	1.00	0.82	0.35	0.85	-0.40	0.09	0.19	0.00	-0.27	0.01	-0.03
4	0.76	0.61	0.57	0.82	1.00	0.17	0.69	-0.51	-0.07	-0.19	-0.20	-0.47	-0.17	-0.17
5	0.43	0.12	0.56	0.35	0.17	1.00	0.57	-0.77	-0.42	0.32	-0.03	-0.52	-0.67	-0.70
6	0.83	0.79	0.86	0.85	0.69	0.57	1.00	-0.49	-0.20	0.13	-0.22	-0.36	-0.18	-0.26
7	-0.35	-0.09	-0.52	-0.40	-0.51	-0.77	-0.49	1.00	0.54	-0.03	0.02	0.79	0.85	0.86
8	0.16	0.27	-0.15	0.09	-0.07	-0.42	-0.20	0.54	1.00	0.62	0.68	0.80	0.68	0.77
9	0.24	0.30	0.28	0.19	-0.19	0.32	0.13	-0.03	0.62	1.00	0.84	0.49	0.12	0.17
10	-0.03	0.10	0.04	0.00	-0.20	-0.03	-0.22	0.02	0.68	0.84	1.00	0.48	0.12	0.22
11	-0.22	0.07	-0.32	-0.27	-0.47	-0.52	-0.36	0.79	0.80	0.49	0.48	1.00	0.84	0.85
12	0.03	0.31	-0.14	0.01	-0.17	-0.67	-0.18	0.85	0.68	0.12	0.12	0.84	1.00	0.98
13	0.01	0.24	-0.22	-0.03	-0.17	-0.70	-0.26	0.86	0.77	0.17	0.22	0.85	0.98	1.00